# Assignment-4.5

TASK-1

Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.

**Step 1: Connect to the Same Network**

- Ensure both your laptop and your friend's phone are connected to the exact same Wi-Fi network (SSID).

**Step 2: Find the Phone's IP Address**

You need the destination address to send the ping. On the Android phone:

1. Go to Settings > About Phone > Status Information (or Network/Wi-Fi settings depending on the brand).
2. Look for IP address (it will look something like 192.168.1.X).

**Step 3: Run the Ping Command from Your Laptop**

If you are using a Windows laptop:

1. Press the Windows Key + R, type cmd, and press Enter to open the Command Prompt.
2. Type ping [Your_Friend's_Phone_IP] (for example: ping 192.168.1.15) and press Enter.
3. Success: If you see replies showing time in milliseconds (Reply from... bytes=32 time=12ms), the devices are successfully communicating!
4. Failure**:** If you get a Request timed out or Destination host unreachable, your router's security features (like AP/Isolate mode) might be blocking devices from talking to each other on the local network.

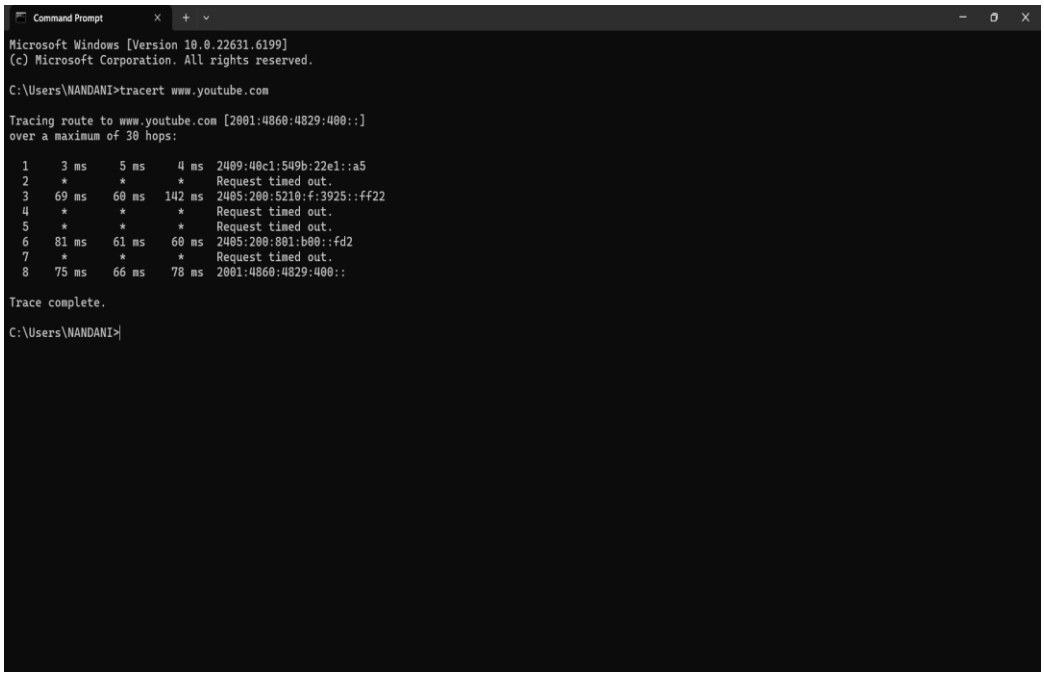
TASK-2

Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

**Windows:** Open the Command Prompt (press Win + R, type cmd, and press Enter), then run:

cmd

tracert www.youtube.com




**Mac or Linux:** Open the Terminal application, then run:

bash

traceroute www.youtube.com

- **Hop Number:** The first column lists the sequential step number (starting from 1 for your local gateway/router).
- **Round-Trip Times (RTT):** The next columns show the time in milliseconds (ms) it takes for packets to reach that specific router and return.
- **IP Address / Hostname:** Identifies the name and network address of each router handling your traffic along the path.

- **Asterisks (*):** Indicate a timeout or a router that is configured not to reply to ICMP/UDP probes (which is common and does not always mean a broken connection).

TASK-3

Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

1. Connect your laptop and your friend's phone to the same Wi-Fi network.
2. Note the phone's IP address.
3. Disconnect the phone from Wi-Fi.
4. On your Windows laptop, open Command Prompt and type:

ping PHONE_IP_ADDRESS

Replace PHONE_IP_ADDRESS with the phone's actual IP address.

Expected observation: You may receive the following message:

**Request timed out.**

This means the laptop did not receive a reply from the phone. You might also see Destination host unreachable, depending on your network configuration.

TASK-4

Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

Observation Table

| IP Address | Device |
|---|---|
| 10.141.249.240 | My Asus VivoBook laptop (Wi-Fi interface) |
| 10.141.249.123 | Likely the hotspot gateway |
| Not detected | Friend's phone — not listed in this output |

TASK-5

Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

**1. Aim**

To set up a network connection, identify connected devices, and test network connectivity using the ping, tracert, and arp -a commands.

**2. Network Setup**

I used my Asus VivoBook laptop running Windows 11 and connected it to a mobile Wi-Fi hotspot. The network connection was checked using Windows Settings and the ipconfig command in Command Prompt. I also planned to connect a friend's mobile phone to the same hotspot to test communication between the devices.

**3. Connectivity Testing Using Ping**

I opened Command Prompt and used the ping command followed by the target device's IP address. This command sends packets to the target and checks whether replies are received.

**Command used:**
ping [Target IP Address]

A successful test displays reply messages, while a failed test may display “Request timed out” or “Destination host unreachable.” The actual result depends on the network connection and whether the target device allows ping requests.

**4. Traceroute Testing**

I used the following command to trace the network path to YouTube:

tracert www.youtube.com

The command displays the network hops between my laptop and the destination. Some hops may show asterisks if routers do not respond to the probes.

**5. Identifying Devices Using ARP**

I ran the arp -a command to view the IP-to-MAC address mappings stored by my laptop. The output showed the Wi-Fi interface address 10.141.249.240 and a dynamic entry for the gateway 10.141.249.123. The friend's phone did not appear in the displayed ARP entries.

**6. Issues Observed**

The ARP table did not identify every connected device. A phone may not appear unless the laptop has recently communicated with it. Ping tests may also fail because of device isolation or firewall settings. These results need to be checked before concluding that a device is unreachable.